ZICHEN LIU

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EDUCATION

Arizona State University — Tempe, AZ
Ph.D. Student in Computer Science — 2023 - present
- **Research area: Software Engineering, Program Analysis, Machine Learning**
- *Advisor: Prof. Xusheng Xiao*

University of Washington — Seattle, WA
Master of Science in Electrical & Computer Engineering — 2019 - 2021

University of California, Berkeley — Berkeley, CA
Exchange Student — 2017

Cuiying Honors College, Lanzhou University — Lanzhou, China
Bachelor of Science in Physics, with highest Honors — 2015 - 2019

PUBLICATIONS

1. **Zichen Liu** and Xusheng Xiao. "AppBDS: LLM-Powered Description Synthesis for Sensitive Behaviors in Mobile Apps." *Proceedings of the 40th IEEE/ACM International Conference on Automated Software Engineering (ASE)*, 2025.
2. **Zichen Liu**, Shao Yang, Xusheng Xiao. "BinCtx: Multi-Modal Representation Learning for Robust Android App Behavior Detection." *arXiv preprint arXiv:2510.14344*, 2025. [arXiv]
3. Jiongxiao Wang*, **Zichen Liu***, Keun Hee Park, Muhao Chen, and Chaowei Xiao. "Adversarial Demonstration Attacks on Large Language Models." *arXiv preprint arXiv:2305.14950*, 2023. [arXiv]
4. Cheng-Yen Yang*, Aotian Zheng*, **Zichen Liu**, and Jenq-Neng Hwang. "Tracklet-based Multi-camera Multiple People Tracking." In *International Conference on Computer Vision (ICCV) Workshops*, 2021. **Ranked 3rd** out of ~80 competitors. [Presentation]

RESEARCH EXPERIENCE

Research Assistant | RISE Lab @ASU
Advisor: Prof. Xusheng Xiao — 2023.7 - present

AppBDS*: LLM-Powered Description Synthesis for Sensitive Behaviors in Mobile Apps*

Research Summary: Mobile apps increasingly access sensitive user data through dangerous permissions (camera, location, etc.), but only 37% adequately explain how this sensitive data is actually used. AppBDSaddresses this transparency gap by combining program analysis with LLMs to automatically generate detailed descriptions of sensitive permission, helping users understand privacy implications beyond basic permission requests.

- Developed AppBDS, a novel framework integrating program analysis with LLMs to generate comprehensive descriptions for Android app permission usage, achieving significant improvements in richness, specificity, and semantic relatedness over state-of-the-art approaches
- Designed UI-Fused Call Graph construction technique combining static analysis with Android UI contexts, reducing analysis complexity from 46,282 to 43 privacy-relevant nodes through intelligent filtering patterns
- Built comprehensive PP-KB knowledge base with 3,500 apps across 7 permission types, implementing multi-agent LLM architecture for cross-validation and evidence-based description synthesis
- Evaluated framework robustness on 270 real-world apps, demonstrating high performance across multiple LLMs (GPT-4o, LLaMA, Qwen) and resilience against code obfuscation

Research Experience

Research Assistant | ASU

Advisor: Prof. Chaowei Xiao 2023.2 - 2023.7

Adversarial Demonstration Attacks on Large Language Models

Research Summary: This research pioneers the study of "Adversarial Demonstration Attacks," a novel threat model targeting the in-context learning (ICL) capability of Large Language Models. By subtly manipulating demonstration examples, we reveal a critical vulnerability that can mislead LLMs into making targeted misclassifications.

- Pioneered the analysis of a novel threat model for LLM in-context learning, allowing attacks via demonstration modifications
- Achieved high attack success on leading LLMs, including GPT2-XL and Llama, across multiple tasks
- Designed experimental methods to evaluate the quality of attacked samples through linguistic/semantic analysis, model perplexity, and user analysis. Highlighting the stealthiness of the attack

Research Assistant | Information Processing Lab @UW

Advisor: Prof. Jenq-Neng Hwang 2020.12 - 2021.9

Multi-camera Multiple People Tracking

- Developed a multi-camera tracking system by integrating a FairMOT baseline with a novel cross-camera tracklet association algorithm.
- Achieved **3rd place (Track 2) and 5th place (Track 1)** in the AICITY Challenge at the ICCV 2021 Workshop, competing against over 70 teams.

Pose-Flow Multi-Object Tracking and Human Pose Estimation

- Proposed **Joint-Flow**, a novel tri-branch architecture that fuses top-down, bottom-up, and temporal flow information for robust pose tracking.
- Attained strong benchmark results on the PoseTrack dataset, achieving **78.7 mAP** and **63.6 MOTA**.

Research Student | Electronic Materials Lab @LZU

Advisor: Prof. Deyan He 2018.12 - 2019.5

Application of MOF based metal oxide & sulfide in energy storage

- Explored a three-in-one method using Zeolitic Imidazolate Frameworks-67 as the precursor to synthesize oxide, sulfide and carbide as electrode materials, and reported excellent electrochemical performance

Industry Experience

Machine Learning Engineer | Ernst Leitz Labs, Leica Camera AG 2021.09 - 2022.11

- Investigate and develop deep learning based method to replace and optimize mobile camera Image Signal Processor (ISP) pipeline.
- Participate in Leitz Looks Engine project: develop image rendering engine and depth estimation algorithms.
- Integrate various depth estimation algorithms such as RAFT-Stereo and MiDaS onto next generation Qualcomm Snapdragon chipset, and perform speed/power optimizing and performance profiling.

Teaching Experience

Teaching Assistant | Arizona State University

- CSE 464 Software Quality Assurance and Testing Fall 2023 – Fall 2025
- CSE 471 Introduction to Artificial Intelligence, Spring 24

Skills

Languages: `Python`, `Java`, `C/C++`

AI LLM: `PyTorch`, `Hugging Face`, `Fine-tuning`, `LangChain`

Analysis Systems: `Soot/SootUp (Static Analysis)`, `System Verilog`

Tools DevOps: `Git`, `Docker`, `CI/CD`

Awards and Honors

- **Outstanding Thesis Award**, Lanzhou University — 2019
- **Academic Merit Scholarship**, Lanzhou University — 2017
- **Outstanding Student Award**, Lanzhou University — 2016

Projects

Blockchain Consensus Simulator

Area: Time Series Analysis — 2020.10 - 2020.12

- Developed a Python simulator to benchmark consensus protocols (e.g., Conflux, Prism), validating their theoretical performance on throughput and latency.

Adversarial Attacks and Defenses: A Survey

Area: Time Series Analysis — 2020.5 - 2020.6

- Authored a comprehensive survey on adversarial ML, synthesizing and categorizing key attack/defense methodologies and identifying research gaps.

A16-bit Arithmetic Logic Unit (ALU) in 65nm Process

Area: Time Series Analysis — 2019.10 - 2019.12

- Designed and implemented a 16-bit ALU using Virtuoso Cadence, winning the **Micron Design Challenge** for the lowest area design.